Deriving the Multi-Dose Concentration Formula

From a First-Order Differential Equation via an Infinite Geometric Series

Generalised to D Doses per P Hours,
with the Continuous Intravenous Limit

Brian Tenneson
M.S., Applied Data Science
Currently Unaffiliated
email: btenneson2301@baypath.edu

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Former Graduate Student
Bay Path University
Department of Applied Data Science

Abstract

Beginning with the first-order elimination law $c'(t) = k_1 c(t)$, I construct the single-dose concentration profile, superpose n doses to obtain a finite geometric series, and pass to the limit $n \to \infty$. The steady-state peak and trough then follow by evaluating a single closed form at the two endpoints of one dosing cycle, and the finite-n expression reproduces the ManyDoseConcentration function of the underlying Mathematica notebooks exactly. In this version the schedule is generalised from one dose per 24 hours to D doses per P hours: every result of the original derivation carries over with the dosing period P replaced by the inter-dose interval $\Delta = P/D$ and each dose rescaled to strength $1/D$. Letting $D \to \infty$ with $x \to 0$ collapses the sawtooth to a flat plateau and recovers the classical continuous-infusion result $C_\infty^{\mathrm{IV}} = \lambda h/\ln 2 = \lambda/k$, so that constant intravenous dosing appears not as a separate model but as the limit of the same geometric series. The derivation reduces to one structural observation: a sum of exponential tails offset by equal time steps is a geometric series. Anchoring that sum at the most recent dose—so that the observation point stays inside the current cycle—makes the closed form, its limit, and both clinical extremes follow in a few lines of algebra. Throughout, the elimination is written in base two so that the half-life is read directly off every formula.

Keywords: pharmacokinetics, first-order elimination, half-life, single-dose profile, absorption ramp, superposition, geometric series, steady state, accumulation multiplier, peak, trough, fluctuation index, dosing frequency, continuous intravenous infusion, infusion rate, ManyDoseConcentration, closed form, dosing period.

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Symbols and standing conventions

The model uses seven quantities, all in hours except where noted.

Symbol	Meaning
t	time (h)
h	elimination half-life (h)
x	absorption duration (h)
P	dosing period (h, default 24)
D	number of doses administered per period P (default 1)
Δ	inter-dose interval, $\Delta = P/D$ (h)
τ	time since the current dose finished absorbing
λ	infusion rate in the continuous limit, $\lambda = 1/P$ (units/h)

Base-two convention. All elimination is written in the form $2^{-t/h}$ rather than $e^{k_1 t}$. In this form the half-life is read off directly: one half-life ($t = h$) gives $2^{-1} = \frac{1}{2}$, two give $2^{-2} = \frac{1}{4}$, and m give 2^{-m}. Every subsequent formula inherits this base, which is why powers of 2 rather than e appear throughout.

Dose-strength normalisation. Sections 1–6 take $D = 1$, so that a single dose reaches unit concentration at the end of absorption. Section 7 generalises to D doses per period, each of strength $1/D$, so that the total amount delivered per period P is held fixed at one unit as D varies. This normalisation is what makes the continuous limit of Section 7 finite; without it the concentration would grow without bound as $D \to \infty$.

Standing assumption. Absorption is assumed to complete before the next dose arrives, $x \leq \Delta$, so that at most one dose is mid-absorption at any time. For $D = 1$ this is the condition $x \leq P$ of Version 1.

1 The Elimination Law and Its Half-Life Parameterisation

A drug is cleared at a rate proportional to the amount present. This first-order law is the entire physical input to the model.

Definition 1.1 (First-order elimination law)**.** The concentration $c(t)$ obeys

$$c'(t) = k_1\, c(t), \qquad k_1 < 0. \tag{1}$$

Proposition 1.2 (General solution)**.** *Every solution of* (1) *has the form*

$$c(t) = k_2\, e^{k_1 t} \tag{2}$$

for a constant k_2.

Proof. Separating variables in (1) gives $dc/c = k_1\, dt$, and integrating yields $\ln|c| = k_1 t + \text{const}$, hence $c(t) = k_2 e^{k_1 t}$. ☐

Proposition 1.3 (Half-life reparameterisation)**.** *The clinically measured quantity is not* k_1 *but the half-life* h*, defined by* $c(t+h) = \frac{1}{2}c(t)$*. Imposing this on* (2) *gives*

$$e^{k_1 h} = \tfrac{1}{2} \iff k_1 h = -\ln 2 \iff k_1 = -\frac{\ln 2}{h}, \tag{3}$$

and substituting (3) *into* (2)*, using* $e^{-(\ln 2)t/h} = 2^{-t/h}$*, converts the solution to base two:*

$$c(t) = k_2 \cdot 2^{-t/h}. \tag{4}$$

Proof. The condition $c(t+h) = \frac{1}{2}c(t)$ applied to (2) cancels $k_2 e^{k_1 t}$ and leaves $e^{k_1 h} = \frac{1}{2}$, which is (3); the base change is the identity $e^{-(\ln 2)t/h} = 2^{-t/h}$. □

Remark 1.4 (Why base two). In the form $2^{-t/h}$ the half-life is manifest, and this is the only reason the change of base is worth making. Every formula below inherits the base, so powers of 2 replace powers of e from this point on.

2 The Single-Dose Concentration Profile

An oral dose is not absorbed instantaneously. Over an absorption window of length x the drug enters the blood at a constant rate; afterwards (4) governs its decay.

Definition 2.1 (Single-dose profile)**.** Normalising so that one dose reaches unit concentration at $t = x$,

$$c_{\text{single}}(t) = \begin{cases} \dfrac{t}{x}, & 0 < t \le x \quad \text{(absorption)}, \\ 2^{-(t-x)/h}, & t > x \quad \text{(elimination)}. \end{cases} \tag{5}$$

Proposition 2.2 (Boundary conditions)**.** *The profile* (5) *satisfies* $c_{\text{single}}(0) = 0$ *and* $c_{\text{single}}(x) = 1$*, and it is continuous at* $t = x$*.*

Proof. At $t = 0$ the ramp gives $0/x = 0$; at $t = x$ the ramp gives $x/x = 1$ while the elimination branch gives $2^0 = 1$, so both branches agree there. □

Definition 2.3 (Finish-of-absorption time)**.** Doses are given every P hours, so dose k is administered at time $(k-1)P$ and finishes absorbing at

$$t_k = (k-1)P + x. \tag{6}$$

Proposition 2.4 (Residual tail of dose k)**.** *For any* $t > t_k$*, the residual contribution of dose* k *is the elimination branch of* (5) *translated to start at* t_k*:*

$$c_k(t) = 2^{-(t-t_k)/h} = 2^{-\left(t-x-(k-1)P\right)/h}. \tag{7}$$

Each past dose therefore leaves its own exponential tail, and successive tails are offset from one another by exactly one period P.

3 Superposition of n Doses: A Finite Geometric Series

Equation (1) is linear, so contributions add.

Definition 3.1 (n-dose superposition)**.** After n doses, at a time t lying in the elimination phase of the current cycle,

$$C_n(t) = \sum_{k=1}^{n} c_k(t) = \sum_{k=1}^{n} 2^{-\left(t-x-(k-1)P\right)/h}. \tag{8}$$

This sum factors in two equivalent ways; the distinction between them matters.

Proposition 3.2 (Form A — anchored at the first dose)**.** *Splitting the exponent in* (8) *and setting* $j = k - 1$*,*

$$C_n(t) = 2^{-(t-x)/h} \sum_{j=0}^{n-1} \left(2^{P/h}\right)^j = 2^{-(t-x)/h} \cdot \frac{2^{nP/h} - 1}{2^{P/h} - 1}. \tag{9}$$

Here the ratio is $2^{P/h} > 1$: measured from the first dose, later terms are larger. Form A is the expression implemented in the notebooks.

Proof. The exponent splits additively,

$$-\frac{t - x - (k-1)P}{h} = -\frac{t - x}{h} + \frac{(k-1)P}{h},$$

so each summand equals $2^{-(t-x)/h}\left(2^{P/h}\right)^{k-1}$. Re-indexing by $j = k - 1$ gives a geometric sum with ratio $2^{P/h}$, whose value is the standard partial-sum identity. □

Definition 3.3 (Time since the most recent dose)**.** Let τ denote elapsed time since the current dose finished absorbing,

$$\tau = t - t_n = t - x - (n-1)P, \qquad 0 \le \tau \le P - x. \tag{10}$$

Proposition 3.4 (Form B — anchored at the most recent dose)**.** *Re-indexing* (8) *by* $i = n - k$ *(cycles into the past),*

$$C_n(\tau) = \sum_{i=0}^{n-1} 2^{-(\tau + iP)/h} = 2^{-\tau/h} \sum_{i=0}^{n-1} r^i, \qquad r := 2^{-P/h} \in (0, 1), \tag{11}$$

and since $r < 1$ *the standard partial sum applies:*

$$\sum_{i=0}^{n-1} r^i = \frac{1 - r^n}{1 - r} \implies C_n(\tau) = 2^{-\tau/h} \cdot \frac{1 - 2^{-nP/h}}{1 - 2^{-P/h}}. \tag{12}$$

Proposition 3.5 (The two forms agree)**.** *Substituting* $t = (n-1)P + x + \tau$ *into* (9) *and dividing numerator and denominator by* $2^{P/h}$ *recovers* (12) *identically:*

$$2^{-(t-x)/h} \frac{2^{nP/h} - 1}{2^{P/h} - 1} = 2^{-\tau/h} \frac{2^{P/h} - 2^{-(n-1)P/h}}{2^{P/h} - 1} = 2^{-\tau/h} \frac{1 - 2^{-nP/h}}{1 - 2^{-P/h}}. \tag{13}$$

Remark 3.6 (The one algebraic trap). Only Form B has a finite limit as $n \to \infty$, because it holds the observation point τ fixed within the current cycle rather than letting t run away with n. Mixing the prefactor of one form with the ratio of the other is the single algebraic trap in this derivation.

4 The Steady-State Limit

Theorem 4.1 (Convergence to steady state)**.** *Since* $0 < r < 1$ *we have* $r^n = 2^{-nP/h} \to 0$*, so* (12) *converges to*

$$C_\infty(\tau) = \lim_{n \to \infty} C_n(\tau) = \frac{2^{-\tau/h}}{1 - 2^{-P/h}} = \frac{2^{(P-\tau)/h}}{2^{P/h} - 1}. \tag{14}$$

Written out, the limit is a plain geometric sum of the tails of all past doses:

$$C_\infty(\tau) = 2^{-\tau/h} + 2^{-(\tau+P)/h} + 2^{-(\tau+2P)/h} + \cdots = 2^{-\tau/h}\left(1 + 2^{-P/h} + 2^{-2P/h} + \cdots\right) = \frac{2^{-\tau/h}}{1 - 2^{-P/h}}. \quad (15)$$

Definition 4.2 (Accumulation multiplier)**.** The factor

$$\left(1 - 2^{-P/h}\right)^{-1} = \frac{2^{P/h}}{2^{P/h} - 1}$$

measures how much steady-state dosing amplifies a single dose. Its two limiting regimes are:

- $h \gg P$ (slow clearance): $2^{P/h} \approx 1 + \dfrac{P\ln 2}{h}$, so the multiplier is $\approx \dfrac{h}{P\ln 2} \gg 1$ — strong accumulation.
- $h \ll P$ (fast clearance): $2^{P/h} \gg 1$ and the multiplier tends to 1 — each dose has essentially cleared before the next.

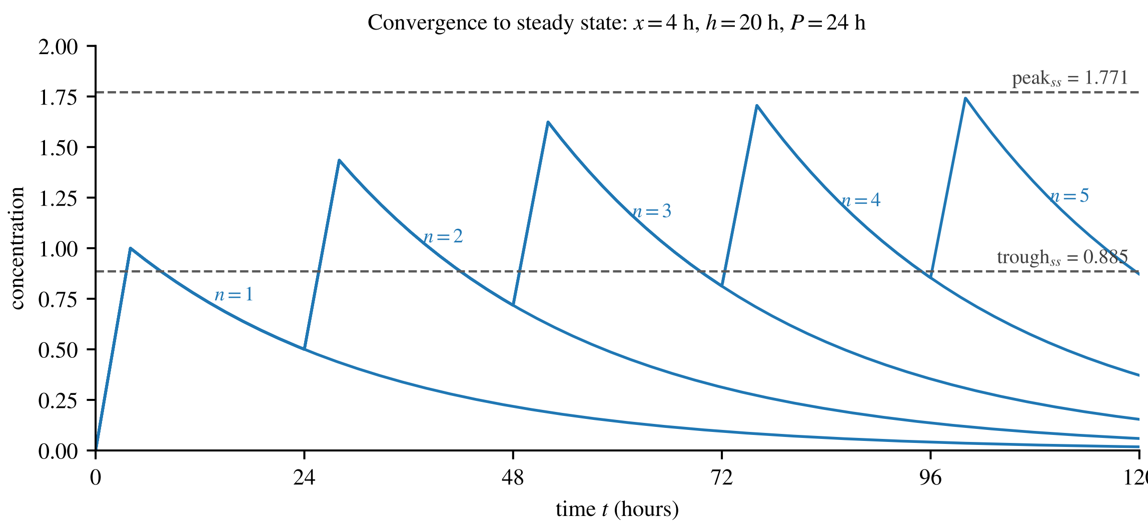

Figure 1: Successive partial sums C_n for $x = 4\,\text{h}$, $h = 20\,\text{h}$, $P = 24\,\text{h}$. Each curve adds one dose; the profile converges to a periodic steady state bounded by the horizontal lines $\text{peak}_{\text{ss}} = 1.771$ and $\text{trough}_{\text{ss}} = 0.885$ derived in Section 5.

5 Peak, Trough, and the Fluctuation Index

Formula (14) decreases monotonically in τ, so the extremes of a steady-state cycle occur at the two endpoints of $\tau \in [0,\, P - x]$.

Theorem 5.1 (Steady-state peak)**.** *At $\tau = 0$ the current dose has just reached unit concentration and all earlier tails are at their largest:*

$$\text{peak}_{\text{ss}} = C_\infty(0) = \frac{1}{1 - 2^{-P/h}} = \frac{2^{P/h}}{2^{P/h} - 1} = 1 + \frac{1}{2^{P/h} - 1}. \quad (16)$$

The final form separates the two physical contributions: the 1 is the dose just absorbed, and $\left(2^{P/h} - 1\right)^{-1}$ is everything left over from all previous doses.

Theorem 5.2 (Steady-state trough)**.** *The minimum falls the instant before the next dose, i.e.* $\tau = P - x$*:*

$$\text{trough}_{\text{ss}} = C_\infty(P - x) = \frac{2^{-(P-x)/h}}{1 - 2^{-P/h}} = \frac{2^{x/h}}{2^{P/h} - 1}, \tag{17}$$

the last step following on multiplying numerator and denominator by $2^{P/h}$.

No approximation is involved: the $2^{x/h}$ arises purely from the offset between the end of absorption and the next dose.

Corollary 5.3 (Fluctuation index)**.** *Dividing* (16) *by* (17)*, the shared denominator cancels and the ratio collapses to a single exponential:*

$$\frac{\text{peak}_{\text{ss}}}{\text{trough}_{\text{ss}}} = \frac{2^{P/h}}{2^{x/h}} = 2^{(P-x)/h}. \tag{18}$$

The swing between peak and trough depends only on how many half-lives fit into the gap $P - x$ between the end of absorption and the next dose. Extending the release window (larger x) or shortening the interval (smaller P) flattens the profile; a short half-life sharpens it.

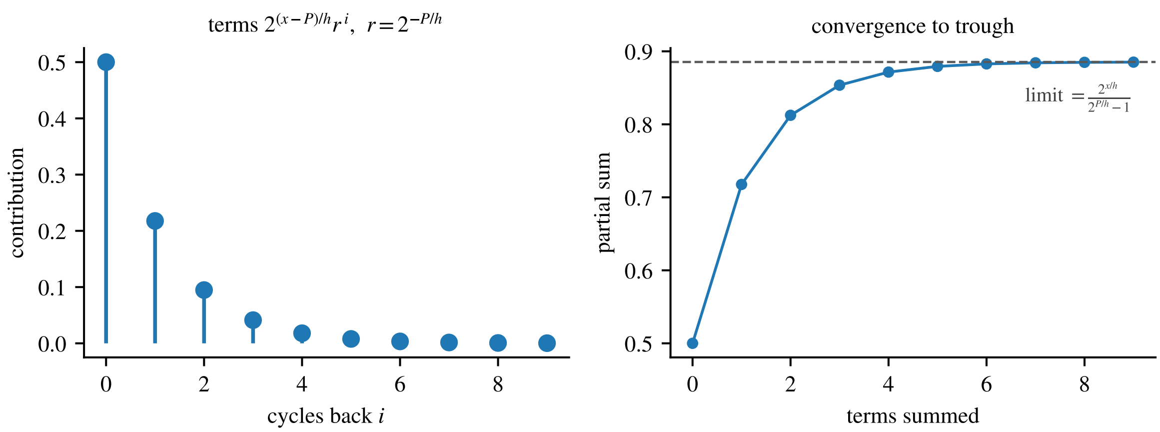

Figure 2: Left: the contributions entering the trough form a geometric sequence with ratio $r = 2^{-P/h}$ and first term $2^{(x-P)/h}$. Right: their partial sums converge to the closed form $2^{x/h}/(2^{P/h} - 1) = \text{trough}_{\text{ss}}$.

Proposition 5.4 (Agreement with the notebook definitions)**.** *With* $P = 24$*, the notebooks define*

$$\texttt{troughmin}[x, h] = \frac{2^{x/h}}{2^{24/h} - 1}, \tag{19}$$

$$\texttt{peakmax}[x, h] = \frac{1}{2^{24/h} - 1} + \frac{\big(12 + x - 12\lceil x/24 \rceil\big)\lceil x/24 \rceil}{x}. \tag{20}$$

Equation (19) *is exactly* (17)*. For* (20)*, whenever* $0 < x \le 24$ *one has* $\lceil x/24 \rceil = 1$*, so the second term reduces to*

$$\frac{(12 + x - 12) \cdot 1}{x} = \frac{x}{x} = 1,$$

and (20) *becomes* $1 + \big(2^{24/h} - 1\big)^{-1}$*, precisely* (16).

The ceiling expression is therefore not an extra physical effect but the generalisation that keeps the formula valid when the absorption window spans more than one dosing period, $x > P$.

6 Recovering the ManyDoseConcentration Function

For finitely many doses the notebooks evaluate Form A, (9), using ceilings to identify the current cycle.

Definition 6.1 (Cycle counters)**.** With $P = 24$, set

$$n = \lceil t/24 \rceil, \qquad m = \lceil (t-x)/24 \rceil, \tag{21}$$

so that n counts doses administered and m counts those that have finished absorbing.

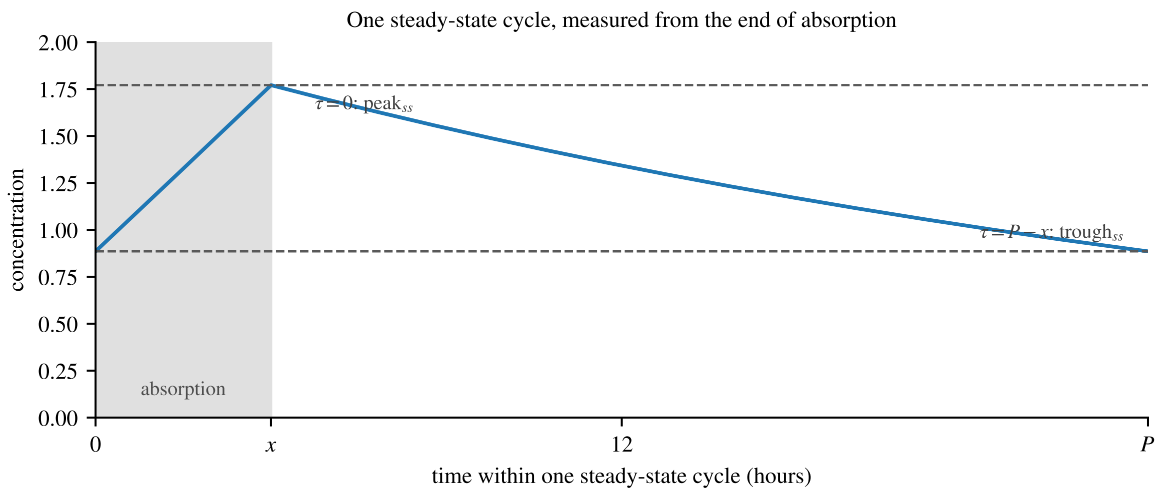

Figure 3: A single steady-state cycle. The maximum occurs the instant absorption ends ($\tau = 0$) and the minimum the instant before the next dose ($\tau = P - x$).

Theorem 6.2 (The complete finite-n profile)**.** *The complete concentration profile is*

$$C(t;x,h) = \begin{cases} \dfrac{(12 + t - 12n)\,n}{x}, & 0 < t \le x, \\[2ex] \underbrace{2^{(x-t)/h}\,\dfrac{2^{24m/h}-1}{2^{24/h}-1}}_{\text{Form A, eq. (9)}} - \underbrace{\dfrac{(n-m)\big(-12 - t + 12n + 12m\big)}{x}}_{\text{absorption-ramp correction}}, & t > x. \end{cases} \tag{22}$$

Letting $m \to \infty$ with τ fixed returns (14)*, the steady state.*

Remark 6.3 (Where the series sits in the code). The first bracket of the lower branch is (9) verbatim, with nP replaced by $24m$: it is the finite geometric series summing the tails of the m doses that have completed absorption. The second bracket subtracts the portion of the linear absorption ramps already accounted for, and vanishes whenever $n = m$ — that is, whenever no dose is mid-absorption.

7 D Doses per P Hours and the Continuous Intravenous Limit

Everything above fixed $D = 1$: one dose per period. Nothing in the derivation used that value, and this section shows that the entire chain (1)–(18) survives the generalisation, with the continuous-infusion formula emerging as a limit rather than as a new model.

7.1 The general schedule

Definition 7.1 (Inter-dose interval and dose strength)**.** Let $D \geq 1$ doses be administered at equal spacing within each period of P hours. The inter-dose interval is

$$\Delta \coloneqq \frac{P}{D}, \tag{23}$$

and each dose carries strength $1/D$, so that the total amount delivered per period P is one unit independently of D. Dose k is administered at time $(k-1)\Delta$ and finishes absorbing at $t_k = (k-1)\Delta + x$, and the standing assumption $x \leq \Delta$ holds.

Theorem 7.2 (General steady state)**.** *Under the schedule of* (23)*, the steady-state concentration at time τ after the current dose has finished absorbing is*

$$C_\infty(\tau) = \frac{1}{D} \cdot \frac{2^{-\tau/h}}{1 - 2^{-\Delta/h}} = \frac{1}{D} \cdot \frac{2^{(\Delta-\tau)/h}}{2^{\Delta/h} - 1}, \qquad 0 \leq \tau \leq \Delta - x. \tag{24}$$

Proof. The argument of Sections 2–4 applies verbatim with P replaced by Δ and each tail (7) multiplied by $1/D$. Explicitly, re-indexing by cycles into the past as in (11),

$$C_n(\tau) = \frac{1}{D} \sum_{i=0}^{n-1} 2^{-(\tau + i\Delta)/h} = \frac{1}{D} 2^{-\tau/h} \sum_{i=0}^{n-1} \rho^i, \qquad \rho \coloneqq 2^{-\Delta/h} \in (0, 1),$$

and since $\rho < 1$ the partial sums converge to $(1-\rho)^{-1}$ as $n \to \infty$, giving (24). The restriction $\tau \leq \Delta - x$ is the standing assumption that no dose is mid-absorption on the interval considered. □

Corollary 7.3 (General peak, trough, and fluctuation index)**.** *Evaluating* (24) *at the two endpoints $\tau = 0$ and $\tau = \Delta - x$,*

$$\mathrm{peak}_{\mathrm{ss}} = \frac{1}{D} \cdot \frac{2^{\Delta/h}}{2^{\Delta/h} - 1} = \frac{1}{D}\left(1 + \frac{1}{2^{\Delta/h} - 1}\right), \tag{25}$$

$$\mathrm{trough}_{\mathrm{ss}} = \frac{1}{D} \cdot \frac{2^{x/h}}{2^{\Delta/h} - 1}, \tag{26}$$

$$\frac{\mathrm{peak}_{\mathrm{ss}}}{\mathrm{trough}_{\mathrm{ss}}} = 2^{(\Delta - x)/h}. \tag{27}$$

Remark 7.4 (Consistency with Version 1). Setting $D = 1$ gives $\Delta = P$ and dose strength 1, so (24)–(27) reduce to (14), (16), (17) and (18) exactly. For $x = 4\,\mathrm{h}$, $h = 20\,\mathrm{h}$, $P = 24\,\mathrm{h}$ this returns $\mathrm{peak}_{\mathrm{ss}} = 1.771$ and $\mathrm{trough}_{\mathrm{ss}} = 0.885$, the values marked in Figure 1.

Remark 7.5 (Effect of dividing the dose). The dose-normalised fluctuation index (27) depends on D only through $\Delta = P/D$, and $2^{(\Delta-x)/h}$ decreases toward 1 as D grows. Splitting the same daily amount into more frequent, smaller doses therefore flattens the concentration profile without altering the average level — the standard clinical rationale for divided dosing and for extended-release formulations.

7.2 The continuous limit

Definition 7.6 (Infusion rate)**.** Since one unit of drug is delivered per period P regardless of D, the time-averaged delivery rate is

$$\lambda := \frac{1}{P} \qquad \text{(units per hour)}, \tag{28}$$

and this quantity is held fixed throughout the limit $D \to \infty$.

Theorem 7.7 (Continuous intravenous limit)**.** *Let $x \to 0$ (instantaneous absorption) and $D \to \infty$ with P fixed, so that $\Delta = P/D \to 0$. Then peak and trough coalesce to the common value*

$$C_\infty^{\mathrm{IV}} \;=\; \lim_{D\to\infty} \frac{1}{D\big(2^{P/(Dh)}-1\big)} \;=\; \frac{h}{P\ln 2} \;=\; \frac{\lambda h}{\ln 2} \;=\; \frac{\lambda}{k}, \qquad k := \frac{\ln 2}{h}. \tag{29}$$

Proof. Setting $x = 0$ in (26) and substituting $\Delta = P/D$,

$$\mathrm{trough}_{\mathrm{ss}} = \frac{1}{D} \cdot \frac{1}{2^{P/(Dh)}-1},$$

which is the expression inside the limit. Writing $2^u = e^{u\ln 2}$ with $u = P/(Dh) \to 0$ and expanding,

$$2^{P/(Dh)} - 1 = e^{(P\ln 2)/(Dh)} - 1 = \frac{P\ln 2}{Dh} + O\left(\frac{1}{D^2}\right). \tag{30}$$

Multiplying (30) by D cancels the factor D and leaves $P\ln 2/h + O(D^{-1})$, whose reciprocal tends to $h/(P\ln 2)$. The remaining equalities are (28) and the definition of k in (3), since $k = -k_1 = \ln 2/h$. That the peak tends to the same value follows from (31) below. $\square$

Corollary 7.8 (The sawtooth collapses)**.** *With $x \to 0$, the fluctuation index* (27) *satisfies*

$$\frac{\mathrm{peak}_{\mathrm{ss}}}{\mathrm{trough}_{\mathrm{ss}}} = 2^{\Delta/h} = 2^{P/(Dh)} \xrightarrow[D\to\infty]{} 2^0 = 1. \tag{31}$$

Peak and trough coalesce, so the steady state is the flat plateau (29)*.*

Remark 7.9 (Equivalent routes to the same limit). The limit may be reached in either of two ways, because D and P enter (29) only through the ratio $\lambda = 1/P$ and the interval $\Delta = P/D$:

- hold $P = 24$ and let $D \to \infty$, as above; or
- hold $D = 1$ and let $P \to 0$, in which case $\mathrm{trough}_{\mathrm{ss}} = 1/(2^{P/h}-1) \to h/(P\ln 2)$ by the same expansion (30).

Both describe the same physical situation: drug entering the blood continuously at rate λ rather than in discrete boluses.

Remark 7.10 (Recovering the classical infusion formula). Equation (29) is the textbook steady-state infusion result $C_{\mathrm{ss}} = R_0/(kV)$ for infusion rate R_0, elimination rate constant k and volume of distribution V, written here in the normalised units of this document ($V = 1$, $R_0 = \lambda$). Its appearance as a limit of the geometric series is the point: continuous infusion is not a separate model but the $\Delta \to 0$ endpoint of the discrete one. The rate of approach is $O(1/D)$ by (30); at $D = 24$ (hourly dosing with $P = 24$) the discrete trough is already within about 2% of the plateau for $h = 20\,\mathrm{h}$.

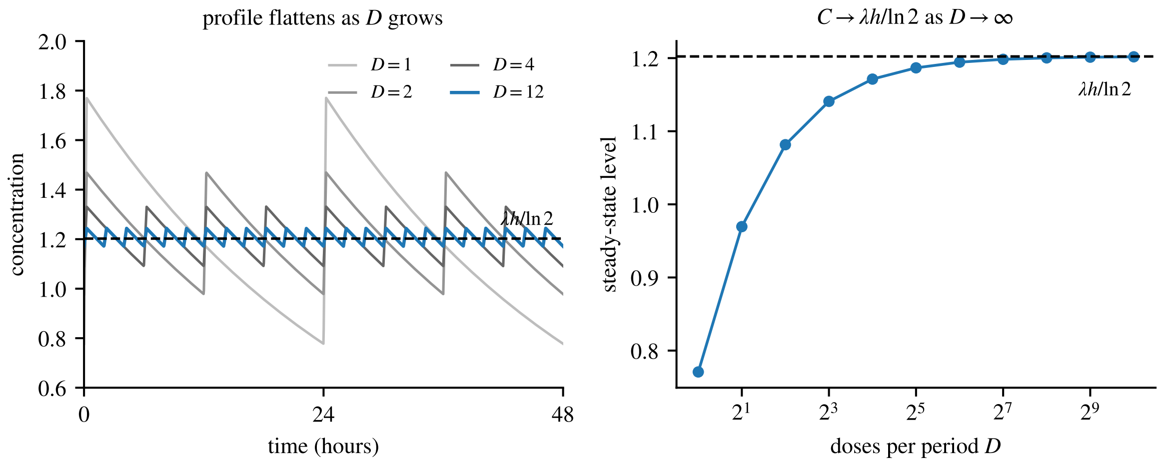

Figure 4: The continuous limit. Left: steady-state profiles for $D = 1, 2, 4, 12$ doses per $P = 24\,\mathrm{h}$ with $h = 20\,\mathrm{h}$ and the total per period held fixed; the sawtooth flattens toward the plateau $\lambda h/\ln 2$ as D grows. Right: the steady-state level $1/\big[D(2^{P/(Dh)} - 1)\big]$ against D, converging to the same plateau at the $O(1/D)$ rate predicted by (30).

8 Summary and Conclusion

The derivation reduces to one observation: a sum of exponential tails offset by equal time steps is a geometric series. Once the sum is anchored at the most recent dose—so that the observation point stays inside the current cycle—the closed form, its limit, and both clinical extremes follow in a few lines of algebra. The base-two convention makes the half-life visible at every stage, and the finite-n ceiling form is nothing more than Form A of the same series with the current cycle identified by $n = \lceil t/24 \rceil$ and $m = \lceil (t - x)/24 \rceil$.

The generalisation of Section 7 costs nothing structurally. Replacing the period P by the inter-dose interval $\Delta = P/D$ and rescaling each dose to strength $1/D$ carries every earlier result across unchanged, and the single additional limit $\Delta \to 0$ turns the geometric series into a constant infusion. Discrete oral dosing and continuous intravenous administration are thus two readings of one formula, separated only by how finely the same total dose is divided in time.

Step	Result	Mathematics
1	Exponential solution	$c' = k_1 c \implies c(t) = k_2\, 2^{-t/h}$
2	Single-dose profile	ramp t/x on $(0, x]$, then $2^{-(t-x)/h}$
3	Superposition of n doses	$C_n(\tau) = 2^{-\tau/h}(1 - r^n)/(1 - r),\ r = 2^{-P/h}$
4	Convergence as $n \to \infty$	$C_\infty(\tau) = 2^{-\tau/h}\big/\big(1 - 2^{-P/h}\big)$
5	Peak, trough, fluctuation	$\dfrac{2^{P/h}}{2^{P/h} - 1},\quad \dfrac{2^{x/h}}{2^{P/h} - 1},\quad 2^{(P-x)/h}$
6	Finite-n form is ManyDoseConcentration	ceilings $n = \lceil t/24 \rceil,\ m = \lceil (t - x)/24 \rceil$
7	D doses per P hours	$\Delta = P/D;\quad C_\infty(\tau) = \dfrac{1}{D} \cdot \dfrac{2^{-\tau/h}}{1 - 2^{-\Delta/h}};$ $\dfrac{\text{peak}_{\text{ss}}}{\text{trough}_{\text{ss}}} = 2^{(\Delta - x)/h}$
8	Continuous IV limit	$D \to \infty, x \to 0$: $C_\infty^{\text{IV}} = \lambda h/\ln 2 = \lambda/k$

Each clinical quantity of interest—the accumulation multiplier, the steady-state peak, the trough, and the fluctuation index—is a one-line consequence of the single closed form (14), the finite-n expression (22) reproduces the ManyDoseConcentration function of the source notebooks exactly, and the continuous-infusion plateau (29) is the $D \to \infty$ limit of the same construction.

Changelog

Version 4

July 26, 2026

Generalised the schedule from one dose per 24 hours to D doses per P hours throughout, introducing the inter-dose interval $\Delta = P/D$ and the dose-strength normalisation $1/D$ (Section 7). Added the general steady state (24) and the general peak, trough and fluctuation index (25)–(27), each verified to reduce to the Version 1 result at $D = 1$. Added the continuous intravenous limit (29), obtained by letting $D \to \infty$ and $x \to 0$, together with the collapse of the fluctuation index (31) and the equivalent route $P \to 0$ at $D = 1$. Added Figure 4 illustrating the approach to the plateau. Added D, Δ and λ to the symbol table, stated the standing assumption $x \le \Delta$, extended the summary table to eight rows, and added this changelog. Running head updated to “D Doses per P Hours”.

Version 1

July 19, 2026

Initial release. Elimination law and half-life reparameterisation, single-dose profile, superposition of n doses as a finite geometric series in two anchorings, the steady-state limit, peak, trough and fluctuation index, and the recovery of the ManyDoseConcentration function with the default schedule of one dose per 24 hours.

Acknowledgements

This document was generated using Fable from Mathematica notebooks created by the author. The source material is the *drugs from scratch* Mathematica notebooks, Levels 1–8 (including the Cymbalta application), together with the Python translation `ManyDoseConcentrationdrugs2.ipynb`. All mathematical claims, interpretations, and final decisions remain the responsibility of the author.

Sources

[1] Tenneson, B. *drugs from scratch* Mathematica notebooks, Levels 1–8 (including the Cymbalta application), and the Python translation `ManyDoseConcentrationdrugs2.ipynb`.

[2] Gibaldi, M., & Perrier, D. (1982). *Pharmacokinetics* (2nd ed.). Marcel Dekker.

[3] Rowland, M., & Tozer, T. N. (2011). *Clinical Pharmacokinetics and Pharmacodynamics: Concepts and Applications* (4th ed.). Lippincott Williams & Wilkins.

